PAPER_299 â Hydrogen Atom UQFF Electrogravitational Dominance Ratio: Î·_EM = 9.65Ã10Â²â¹

Session: 85
Module: HYDROGEN_ATOM_UQFF_MODULE.cpp (27th C++ UQFF module â FIRST atomic-scale module)
System: Hydrogen ground state â Bohr model, M_p = 1.6726Ã10â»Â²â· kg, r_Bohr = 5.2918Ã10â»Â¹Â¹ m, z = 0
Category: Electrogravitational Dominance â FIRST UQFF atomic module, FIRST Î·_EM computation
UQFF Version: 2.0

Abstract

At the atomic scale, the UQFF framework reveals a fundamental asymmetry between gravitational and electromagnetic forces. The Newtonian gravitational acceleration at the Bohr radius (g_base = 3.99Ã10â»Â¹â· m/sÂ²) is the smallest base-gravity value across all 27 UQFF modules â five orders of magnitude below the previous minimum (M16 Eagle Nebula, 1.454Ã10â»Â¹Â² m/sÂ²). The electron Lorentz acceleration in the atomic magnetic field (a_Lorentz = qÃv_orbÃB/m_e = 3.85Ã10Â¹Â³ m/sÂ²) completely dominates the UQFF total. The ratio Î·_EM = a_Lorentz/g_base = **9.65Ã10Â²â¹** defines the UQFF Electrogravitational Dominance Ratio â the largest force asymmetry computed in the UQFF framework.

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1. Physical Setup

The hydrogen atom in its ground state is the smallest gravitational system tractable in the UQFF framework:

Parameter	Value	Units
Proton mass M_p	1.6726Ã10â»Â²â·	kg
Bohr radius r_Bohr	5.2918Ã10â»Â¹Â¹	m
Electron mass m_e	9.1094Ã10â»Â³Â¹	kg
Electron orbital velocity v_orb = Î±Â·c	2.1877Ã10â¶	m/s
Atomic magnetic field B_atom (est.)	1.0Ã10â»â´	T
Electron charge q	1.6022Ã10â»Â¹â¹	C
Fine-structure constant Î±	7.2974Ã10â»Â³	â

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2. Core Equations

2.1 Newtonian Base Gravity

$$g_{\text{base}} = \frac{G \cdot M_p}{r_{\text{Bohr}}^2} = \frac{6.6743 \times 10^{-11} \times 1.6726 \times 10^{-27}}{(5.2918 \times 10^{-11})^2}$$

$$g_{\text{base}} = \frac{1.116 \times 10^{-37}}{2.800 \times 10^{-21}} = 3.986 \times 10^{-17}\ \text{m/s}^2$$

Smallest g_base in all UQFF modules. Prior minimum: M16 Eagle Nebula (1.454Ã10â»Â¹Â² m/sÂ²). Hydrogen is 5 orders of magnitude smaller.

2.2 Electron Lorentz Acceleration (Dominant Term)

$$a_{\text{Lorentz}} = \frac{q \cdot v_{\text{orb}} \cdot B}{m_e} = \frac{1.6022 \times 10^{-19} \times 2.1877 \times 10^{6} \times 10^{-4}}{9.1094 \times 10^{-31}}$$

$$a_{\text{Lorentz}} = \frac{3.504 \times 10^{-17}}{9.109 \times 10^{-31}} = 3.848 \times 10^{13}\ \text{m/s}^2$$

2.3 Electrogravitational Dominance Ratio [PAPER_299]

$$\eta_{\text{EM}} = \frac{a_{\text{Lorentz}}}{g_{\text{base}}} = \frac{3.848 \times 10^{13}}{3.986 \times 10^{-17}} = \mathbf{9.65 \times 10^{29}}$$

The electromagnetic Lorentz force exceeds Newtonian gravity by **30 orders of magnitude** at the hydrogen Bohr radius.

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3. Computed Values

Quantity	Value	Units	Notes
g_base	3.986Ã10â»Â¹â·	m/sÂ²	Newtonian at r_Bohr
a_Lorentz	3.848Ã10Â¹Â³	m/sÂ²	DOMINANT EM term
Î·_EM	**9.65Ã10Â²â¹**	â	**[PAPER_299]**
a_Ug (Ug1+Ug4)	7.972Ã10â»Â¹â·	m/sÂ²	2Ãg_base
a_quantum (HUP)	~1.5Ã10â»Â³â´	m/sÂ²	(Ä§/â(ÎxÂ·Îp))Ã2Ï/t_H
a_osc (Lyman)	~2Ã10â»Â¹â°	m/sÂ²	[PAPER_300]
a_GR_min	~2.8Ã10â»â¶â°	m/sÂ²	[PAPER_301]
a_Lambda	3.30Ã10â»Â³â¶	m/sÂ²	cosmological
Total g_H (t=0)	~3.85Ã10Â¹Â³	m/sÂ²	EM-dominated

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4. UQFF g_base Spectral Catalog

The hydrogen module defines the **EM-dominated regime** and the minimum g_base across all UQFF modules:

Module	System	g_base (m/sÂ²)	Session
Hydrogen (THIS)	H atom, r_Bohr	**3.99Ã10â»Â¹â·**	**85**
M16 Eagle Nebula	nebula ~35 ly	1.454Ã10â»Â¹Â²	80
M87 (hypothetical)	galaxy	~10â»Â¹â°	â
HUDF z=3.5	deep field galaxy	~10â»Â¹â°	72g
Observable Universe	r=4.4Ã10Â²â¶ m	3.45Ã10â»Â¹â°	84
Saturn	planetary	10.44	78

Hydrogen g_base is **5 orders smaller** than the prior minimum (M16) and **27 orders smaller** than Saturn.

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5. Physical Interpretation

The Î·_EM ratio of 9.65Ã10Â²â¹ quantifies the well-known dominance of electromagnetism over gravity at atomic scales. Within the UQFF framework, this is the first direct computation of this asymmetry as a UQFF term ratio. The result is consistent with the known force ratio (electromagnetic/gravitational â 10Â³â¶ for electron-proton, with Î·_EM here representing the Lorentz/Newtonian ratio specifically at v_orb and B_atom=10â»â´ T).

The EM term a_Lorentz represents the centripetal Lorentz force maintaining the electron in its Bohr orbit, expressed as acceleration per unit mass of the system (proton reference frame). This bridges classical orbital mechanics to the UQFF gravitational framework.

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6. UQFF 2.0 Implementation

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// [PAPER_299] in HYDROGEN_ATOM_UQFF_MODULE.cpp updateCache():
g_base_cache     = G_NEWTON * M_proton / (r_Bohr * r_Bohr);    // 3.99e-17 m/s^2
a_Lorentz_cache = Q_ELEM * v_orb * B_atom / m_elec;           // 3.85e13  m/s^2
eta_EM_cache     = a_Lorentz_cache / g_base_cache;              // 9.65e29 [PAPER_299]
```

WOLFRAM_TERM: `HYDROGEN_LORENTZ = "a_Lorentz = q*v_orb*B/m_e = 3.85e13 m/s^2; eta_EM = 9.65e29 [PAPER_299]"`

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7. Significance

1. **First UQFF atomic-scale module**: Establishes the lower bound of the UQFF gravity spectrum
2. **First Î·_EM computation**: Defines the electrogravitational dominance ratio as a named UQFF quantity

3. **Minimum g_base**: 3.99Ã10â»Â¹â· m/sÂ² â 5 orders below prior module minimum (M16)
4. **EM-dominated limit**: All 26 prior modules were gravity-dominated or GR-dominated; hydrogen is the first EM-dominated
5. **Scale anchor**: Provides the atomic-scale end of the UQFF multi-scale framework (companion to Universe-scale PAPER_296â298)

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8. Cross-References

- **PAPER_298** (Session 84): GR Curvature Dominance Îµ_GR = 5.056 > 1 at Universe scale â opposite extreme
- **PAPER_301** (Session 85): Proton Îµ_GR = 7.04Ã10â»â´â´ â GR spectral minimum (same module)
- **PAPER_288** (Session 81): DPM-THz Cascade, RSC module â prior smallest UQFF scale (~10 km plasma)
- **PAPER_266** (Session 72g): Gravitational Meissner Effect â another EM-gravity interface

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9. Summary

$$\boxed{\eta_{\text{EM}} = \frac{a_{\text{Lorentz}}}{g_{\text{base}}} = \frac{q \cdot v_{\text{orb}} \cdot B/m_e}{G \cdot M_p/r_{\text{Bohr}}^2} = 9.65 \times 10^{29}}$$

The hydrogen atom UQFF module establishes the electrogravitational boundary: at the Bohr radius, the Lorentz electromagnetic force exceeds Newtonian gravity by 30 orders of magnitude. This defines the **EM-dominated regime** of the UQFF framework â the atomic-scale complement to the gravity-dominated and GR-dominated regimes of larger UQFF modules.

UQFF computed: UQFF energy correction term [SSq]×h?_g/(k_B×T) = 0.57 × 7.7e-50/(1.38e-23 × 300) = 1.1e-29; UQFF shift in Lyman-alpha = 1.1e-29 × 13.6 eV.